

Functions of many variables – 4

Homogeneity, scaling and functional equations

Homework

11 сентября 2026 года

Theoretical minimum. Euler's operator $E = x_1\partial_1 + \dots + x_n\partial_n$ measures the degree: $Ef = kf$ is equivalent to $f(tx) = t^k f(x)$, and E is a derivation. Substituting tx and comparing powers of t separates homogeneous components; a homogeneous function is determined by its values on the unit sphere. In the coordinates $r = \|x\|$, $\omega = x/\|x\|$ the operator E becomes $r\partial_r$, so an equation $P(E)f = g$ is an ordinary differential equation in $\ln r$ — and a repeated root of P produces a logarithm. Hadamard's lemma divides by a coordinate, $f(x) = \sum x_i \int_0^1 \partial_i f(tx) dt$. A representation $f(x, y) = \sum_{i \leq k} g_i(x) h_i(y)$ exists exactly when the sections of f span a space of dimension at most k . In functional equations of several variables one substitutes coincident values, iterates towards a limit, and describes the orbits of the group of substitutions; continuity is used last.

1. Prove Hadamard's lemma in the form $f(x) = \sum_i x_i g_i(x)$ with $g_i \in C^{k-1}$ for $f \in C^k$ with $f(0) = 0$, and its second-order version: if in addition $\nabla f(0) = 0$, then $f(x) = \sum_{i,j} x_i x_j h_{ij}(x)$ with continuous h_{ij} , which may be taken symmetric.
2. (Proposed problems) Find all functions $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ satisfying $f(xu + yv, -xv + yu) = f(x, y)f(u, v)$ for all $x, y, u, v \in \mathbb{R}$. Which of them are continuous?
3. (ISTCiM 2019) Find all functions $F: \mathbb{R}_+^2 \rightarrow \mathbb{R}_+$ such that

$$F(x, y) = F\left(\frac{x+y}{2}, \frac{2}{\frac{1}{x} + \frac{1}{y}}\right) \quad \text{and} \quad \frac{x+y}{2} \geq F(x, y) \geq \frac{2}{\frac{1}{x} + \frac{1}{y}} \quad \text{for all } x, y > 0.$$

4. (VJIMC 2004) Find all functions $f: \mathbb{R}_0^+ \times \mathbb{R}_0^+ \rightarrow \mathbb{R}_0^+$ such that

- 1) $f(x, 0) = f(0, x) = x$ for all $x \geq 0$;
 - 2) $f(f(x, y), z) = f(x, f(y, z))$ for all $x, y, z \geq 0$;
 - 3) there is a real k with $f(x+y, x+z) = kx + f(y, z)$ for all $x, y, z \geq 0$.
5. (IMC) Call a polynomial $P(x_1, \dots, x_k)$ good if there exist 2×2 real matrices $A_1, \dots, A_k$ with $P(x_1, \dots, x_k) = \det(\sum_{i=1}^k x_i A_i)$. Find all values of k for which every homogeneous polynomial of degree 2 in k variables is good.
6. Let f be continuous on $\mathbb{R}^n \setminus \{0\}$ and homogeneous of degree 0. Prove that f is determined by its restriction to the unit sphere and that it extends continuously to the origin only if it is constant. Then construct a rational function of two variables, homogeneous of degree 0, for which both iterated limits at the origin exist and are equal while the limit does not exist.
7. Prove that a polynomial vanishing on the unit sphere of $\mathbb{R}^n$ is divisible by $x_1^2 + \dots + x_n^2 - 1$, and that the homogeneous components of a harmonic polynomial are harmonic. (The second statement is the first step in problem 1 of the homework of Functions of many variables – 1.)
8. Solve the equation $Ef = kf + g$ on $\mathbb{R}^n \setminus \{0\}$, where g is homogeneous of degree k , and show that the logarithm in the answer cannot be avoided; explain in this language why the solutions in problem 3 of the class sheet contain $\ln x$.
9. Prove that a continuous $f: (0, \infty)^2 \rightarrow \mathbb{R}$ with $f(tx, ty) = t^k f(x, y)$ for all $t > 0$ has the form $f(x, y) = x^k \varphi(y/x)$ with continuous φ , and use this to describe all C^1 solutions of $xf_x + yf_y = kf$ on the quadrant.
10. Prove that a representation $f(x, y) = \sum_{i=1}^k g_i(x)h_i(y)$ exists if and only if every $(k+1) \times (k+1)$ matrix $(f(x_i, y_j))$ is degenerate. Deduce that $\frac{1}{x+y}$ admits no such representation for any k , and compare with problem 3 of Polynomials – 8.